Peter_individual_summary

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Abstract

I would like to divide this project into the following sections: **Data extraction**, **Preprocessing**, **Exploratory Data Analysis**, **Modeling**. We fairly divided each part among everyone to complete. Therefore, all of us participated in the entire project. Next, I will introduce what I specifically did in this project.

Contributions

Data extraction

In the beginning of the project, Wendy and I utilized the MATLAB code provided by Michaela to extract the environmental(CASE-STSE) data from the SCRIPPS website (https://ecco.ucsd.edu/case_stse_results1.html). I modified the MATLAB code and extracted some of the environmental data(**Salinity**, **Zonal_velocity**, **Meridional_velocity**, **Vertical_velocity**) from 2009 to 2023. The MATLAB code is not stored in the GitHub.

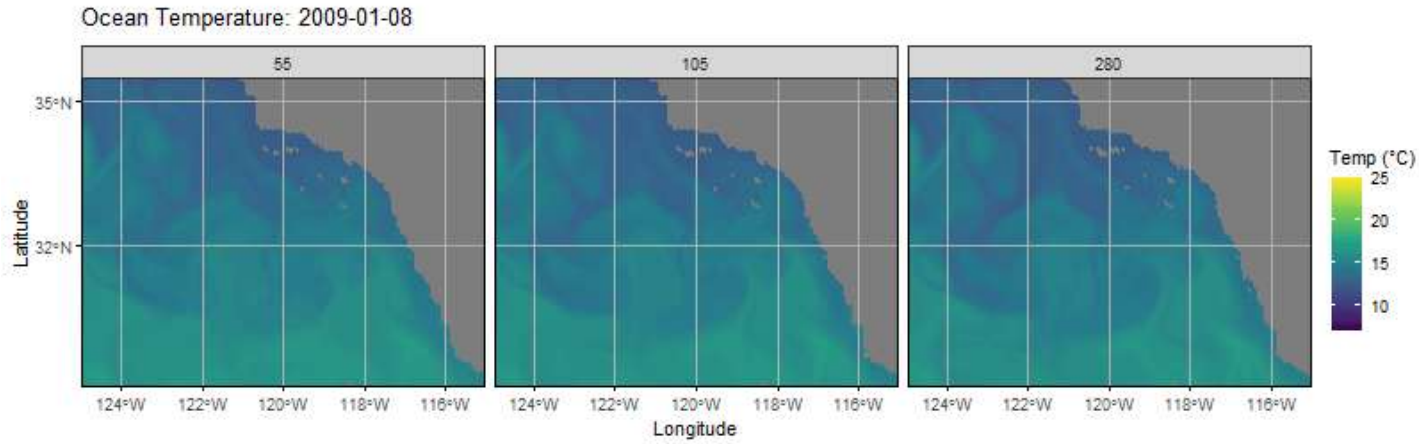
Preprocessing

In this part, I was tried to calculate the **monthly average** of each environmental variable. The script is located at:

```
|- script/enviro_data_average/STSE_eviro_month_avg.R & temperature_average_code.R
```

Exploratory Data Analysis

I worked with Josh to create an **animation** to show how environmental variables change over time by using **ggOceanMaps** and **gganimate** packges.



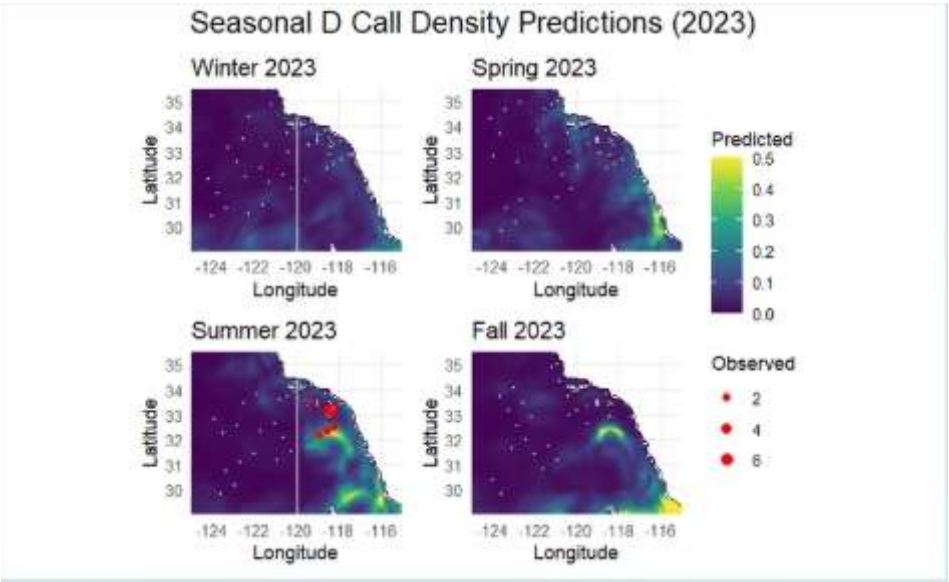
Modeling

I primarily focused on building models for **D call**. I tried different combination of variables and model selections to find the best model and showed its results. The Rmd file is located at:

```
|- script/modeling/d_call/d_call.Rmd
```

Also, I helped Justin to write the function of **prediction heatmap**, which is used to visualize the results of each model prediction. The function is showed in `d_call.Rmd` file.

The prediction heatmap of D call's best model:



Summary

My teammates and I did a really great job in this project. Everyone in this team is nice. I am very happy and thanksful for meeting such a wonderful group of partners and completed such a great project. In addition, I would like to thank Michaela for providing us with this project and Dr. Baracaldo for for her guidance.